have only positive eigenvalues, the harmonic oscillator has no negative energy levels, and in fact, it has no state with zero energy either.

If all the energy levels of a system must be positive, there must be a lowest allowable energy and a wave function to go with it. This lowest energy level is called the *ground state* and is denoted by $\psi_0(x)$. Keep in mind that the subscript 0 does not mean that the energy is zero; it means that it is the lowest allowable energy.

There is a very useful mathematical theorem that helps identify the ground state. We won't prove it here, but it is very simple to state:

> *The ground-state wave function for any potential has no zeros and it's the only energy eigenstate that has no nodes.*

So all we have to do to find the ground state of our harmonic oscillator is to find a nodeless solution for some value of E. It doesn't matter how we find it—we can use mathematical tricks, make guesses, or just ask the professor. Let's use the latter method. (I'll play the role of the professor.)

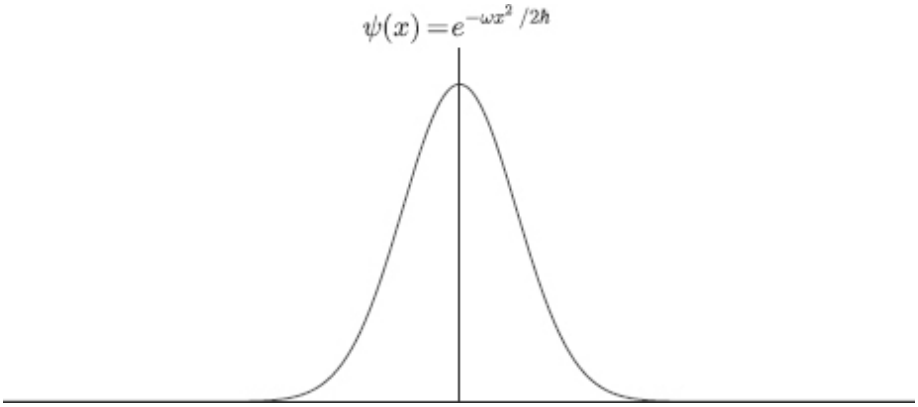


Figure 10.1: Harmonic Oscillator Ground State

Here is a function that works:

$$\psi(x) = e^{-\frac{\omega}{2\hbar}x^2}. \qquad (10.15)$$

This function is shown schematically in Fig. 10.1. As you can see, it's concentrated near the origin, where we expect the lowest energy state to be concentrated. It goes to zero very quickly as it moves away from the origin,